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ORIGINAL ARTICLE

Continuous Glucose Monitor Use Prevents Glycemic Deterioration in Insulin-Treated Patients with Type 2 Diabetes

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Abstract

Continuous glucose monitoring (CGM) is indicated in poorly controlled insulin-treated patients with type 2 diabetes (T2D) to improve glycemic control and reduce the risk of hypoglycemia, but the benefits of CGM for lower risk patients have not been well studied. Among 17,422 insulin-treated patients with T2D with hemoglobin A1c (HbA1c) <8% and no recent severe hypoglycemia (based on emergency room visits or hospitalizations), CGM initiation occurred in 149 patients (17,273 noninitiators served as reference). Changes in HbA1c and severe hypoglycemia rates for the 12 months before and after CGM initiation were calculated. CGM initiation was associated with decreased HbA1c (−0.06%), whereas noninitiation was associated with increased HbA1c (+0.32%); a weighted adjusted difference-in-difference model of change in HbA1c yielded a net benefit of −0.30%; 95% CI −0.50%, −0.10%; $P=0.004$). No significant differences were observed for severe hypoglycemia. CGM may be useful in preventing glycemic deterioration in well-controlled patients with insulin-treated T2D.

Keywords: Continuous glucose monitors, Self-monitoring of blood glucose, Type 2 diabetes, Glycemic control, Hypoglycemia, Real-world evidence.

Introduction

RANDOMIZED CONTROLLED TRIALS have demonstrated the efficacy of continuous glucose monitoring (CGM) to improve glycemia in poorly controlled patients with insulin-treated type 2 diabetes (T2D).^{1,2} We recently reported that initiation of real-time CGM, relative to self-monitoring of blood glucose (SMBG) alone, was associated with improvement in glycemic control and reduction in severe hypoglycemia (based on emergency department [ED] visits or hospitalization) in an observational cohort of 36,080 insulin-treated patients

with T2D.³ However, the benefits of CGM specifically in lower risk patients have not been well studied. We conducted a real-world comparative effectiveness study of the benefits of CGM use in insulin-treated patients with hemoglobin A1c (HbA1c) <8% and no history of severe hypoglycemia.

Materials and Methods

This study cohort included 17,422 insulin-treated T2D patients who used SMBG, had HbA1c <8%, and no recent severe hypoglycemia (ED visits or hospitalizations for

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hypoglycemia). CGM initiation (real-time CGM) was identified among patients during 2015–2019 and the remaining noninitiators served as a reference group. All patients received care from Kaiser Permanente Northern California, a fully integrated health care delivery system providing care for 4.6 million members. A retrospective cohort design was used to estimate changes in HbA1c and severe hypoglycemia rates associated with CGM initiation compared with SMBG use alone (active comparator reference group). The Kaiser Permanente Northern California Institutional Review Board approved this study.

The outcomes of interest were HbA1c and severe hypoglycemia rates (based on a primary diagnosis from an ED visit or a principal diagnosis from a hospitalization). The exposure of interest was CGM initiation. Changes in the outcomes attributable to CGM initiation were calculated as difference-in-differences (DiD) estimates defined as the pre-post changes during the 12 months before and after CGM initiation (exposure) minus the pre-post changes among patients who used SMBG alone (reference group using a randomly assigned date as baseline).⁴ The size and sign of the DiD estimate indicate the magnitude and direction of the net changes in outcome in CGM initiators accounting for changes observed in the reference group.

In addition to the crude DiD estimates, confounding was addressed by specifying adjusted and propensity score-based overlap weighted,^{5,6} repeated measures, generalized linear models, assuming a compound symmetry covariance structure to account for nonindependence of the residual error.⁶ Conservative 95% confidence intervals were calculated using the Huber White sandwich estimator method.⁷ These doubly robust⁸ DiD estimates were intended to balance covariates in the exposed and reference groups and emulate a randomized clinical trial.⁶

To calculate the propensity scores, treatment models were specified to predict the probability of CGM initiation using >40 candidate predictors and all possible interactions and quadratic terms as inputs (see Supplementary Table S1 for list of variables included in the propensity score models) using a machine learning approach to stepwise logistic regression with model decisions based on Akaike information criterion. Instrumental variables (e.g., cost-sharing for CGM) were purposely excluded from the treatment models.⁹ Missing data were rare (~0.4% of the data points). Median values were imputed for missing continuous variables and indicators for missing categorical variables were added to the propensity score models.

The propensity scores were then used to derive overlap weights using the formula (overlap weight = 1 – propensity score) for the CGM initiators and (overlap weight = propensity score) for the noninitiators (reference group).^{5,6,10} Overlap weights give more influence to (i.e., upweight) exposed individuals (i.e., CGM initiators) predicted to have a low likelihood of being exposed and to unexposed individuals (i.e., noninitiators) predicted to have a high likelihood of being exposed.

Covariate differences before and after overlap weighting were assessed using the standardized difference (Cohen's D^{11}), which is the difference in means or proportions divided by pooled standard deviation. D indicates the size and direction of imbalance in the distribution of characteristics in exposed versus reference groups. A positive D for a given character-

istic means patients were overrepresented in CGM initiators and underrepresented in the noninitiators, and vice versa for a negative standardized difference. $|D| < 0.1$ is considered a negligible imbalance.¹² The overlap weights balanced the majority of covariates specified in the treatment model between exposed and the reference groups (i.e., created exchangeability).⁶ Variables that remained unbalanced in the exposed and reference group despite overlap weighting were included in the final statistical model to adjust for the residual covariate imbalance.

Analysis of a directed acyclic graph (DAG)¹³ of the hypothesized causal framework identified potential confounding variables: insulin injection frequency, use of rapid or short-acting insulin, insulin pump use, a history of acute metabolic crises, and HbA1c results 12–24 months before baseline (i.e., assessed in the 12 months before the beginning of the pre-baseline period). The final weighted models were further adjusted for these putative confounders.

We checked for violations of the DiD model assumptions, including Experimental Treatment Assignment,^{14,15} parallel trends,⁴ common shock, or no spillover⁴; none were identified. P -values were two-sided, with a significance threshold of .05. Statistical analyses were performed using SAS version 9.4 (SAS Institute) and R.

Results

In this cohort of 17,422 insulin-treated patients with T2D who used SMBG, had HbA1c <8%, and no recent severe hypoglycemia, 149 patients initiated CGM (Table 1). Compared with noninitiators, CGM initiators were more likely to be younger at baseline, younger at diabetes onset, have fewer comorbidities, and have a higher predicted risk of hypoglycemia; CGM initiators were less likely to be Hispanic or black, have limited English-proficiency, or live in a deprived neighborhood.

In addition, compared with noninitiators, CGM users were more likely to be using basal-bolus insulin, long-acting analog insulin (vs. Neutral Protamine Hagedorn [NPH]), rapid-acting analog insulin (vs. regular), insulin pens, or insulin pumps, and they were more likely to have been dispensed glucagon; CGM users were less likely to be using mixed insulin, multiple daily injections, sulfonylureas, or ≥ 2 therapeutic medication classes. After overlap weighting, most of the differences in baseline characteristics were balanced ($|D| < 0.1$) between CGM initiators and noninitiators. To correct for the residual imbalance, the few baseline characteristics that remained unbalanced ($|D| \geq 0.1$) were added as covariates in the DiD models along with the critical confounders identified from the DAG analysis.

Mean HbA1c declined among initiators from 6.98% to 6.92% (difference -0.06%) and increased among noninitiators from 7.10% to 7.42% (difference $+0.32\%$), yielding a crude (unweighted and unadjusted) DiD estimate of -0.37 (-0.50 , -0.24) and a weighted, adjusted DiD estimate of -0.30% ; 95% CI -0.50% , -0.10% ; $P=0.004$ (Table 2). Changes in HbA1c were also estimated as the proportion of participants having HbA1c <7%, <8%, or >9% (consistent with Health care Effectiveness Data and Information Set (HEDIS) performance measures).

The prevalence of HbA1c <7% increased from 40.3% to 53.7% (difference $+13.4\%$) among CGM initiators compared

TABLE 1. BASELINE CHARACTERISTICS BEFORE AND AFTER PROPENSITY SCORE OVERLAP WEIGHTING IN 17,422 INSULIN-TREATED PATIENTS WITH TYPE 2 DIABETES WHO USED SMBG AND HAD HbA1c <8% AND NO RECENT SEVERE HYPOGLYCEMIA WHO WERE REAL-TIME CONTINUOUS GLUCOSE MONITORING INITIATORS (N=149) OR NONINITIATORS (N=17,273) DURING JANUARY 1, 2015–DECEMBER 31, 2019

Variable	Crude (unweighted) ^a			After overlap weighting ^{a,b}		
	CGM initiators N=149	Noninitiators N=17,273	Standardized difference ^c	CGM initiators N=149	Noninitiators N=17,273	Standardized difference ^c
Sex						
Male	53.0	50.6	0.05	51.2	48.4	0.06
Female	47.0	49.4	−0.05	48.9	51.6	−0.06
Age (years), mean (SD)	61.0 (13.4)	66.3 (11.5)	−0.42	61.1 (13.9)	62.3 (12.6)	−0.09
Age at DM onset, mean (SD)	42.4 (12.5)	49.7 (11.0)	−0.62	42.4 (12.6)	44.0 (12.1)	−0.13
DM duration in years, mean (SD)	18.4 (12.8)	16.1 (8.8)	0.21	18.6 (12.8)	17.1 (10.0)	0.13
Race/ethnicity ^d						
White	65.8	49.4	0.34	61.4	58.7	0.05
Hispanic	5.4	18.7	−0.42	5.5	10.7	−0.19
Black	6.7	9.4	−0.10	7.1	6.9	0.01
Asian	18.1	16.4	0.05	20.4	18.0	0.06
Mixed	2.7	4.0	−0.07	3.7	3.8	−0.01
Other	0	1.5	−0.17	0	0.9	−0.13
Unknown	1.3	0.6	0.07	1.9	1.0	0.08
Preferred spoken language was other than English ^e	1.3	7.7	−0.31	1.9	2.6	−0.05
Neighborhood deprivation index ^c						
Q1 (least deprived)	34.9	20.1	0.34	35.1	30.0	0.11
Q2	28.9	28.7	0.00	28.8	30.7	−0.04
Q3	24.8	29.2	−0.10	22.7	23.0	−0.01
Q4 (most deprived)	10.7	21.3	−0.29	12.5	15.6	−0.09
Insurance type ^f						
Commercial	96.6	97.3	−0.04	96.5	96.7	−0.01
Medicare advantage	55.0	63.6	−0.17	54.2	55.2	−0.02
Medicaid	6.0	4.5	0.07	5.2	5.4	−0.01
Charlson comorbidity score ^g						
1	22.2	19.8	0.06	19.4	18.0	0.04
2	28.2	20.2	0.19	26.3	26.9	−0.01
3	10.7	10.1	0.02	11.2	12.5	−0.04
4+	38.9	49.9	−0.22	43.1	42.7	0.01
Hypoglycemia risk score ^h						
Low	69.1	68.3	0.02	65.9	70.3	−0.09
Intermediate	10.7	25.0	−0.38	11.6	18.4	−0.19
High	20.1	6.7	0.40	22.5	11.3	0.30
Insulin type						
Basal						
Long (analog)	53.7	13.1	0.95	52.6	52.4	0
NPH ⁱ	43.6	75.4	−0.68	53.4	41.8	0.23
Bolus						
Rapid (analog)	81.2	13.4	1.85	74.2	71.0	0.07
Short	18.8	29.2	−0.25	23.1	21.3	0.04
Mixed	2.7	12.3	−0.37	3.4	4.1	−0.04
Glucagon dispensed during prior 12 months ^j	23.5	1.1	0.73	15.5	13.8	0.05
Insulin pen user	41.6	13.9	0.65	51.2	45.5	0.11
Insulin pump user ^k	28.2	0.1	0.88	9.4	8.9	0.02
Multiple daily insulin injections (≥3 times/day)	16.8	24.3	−0.19	21.6	22.9	−0.03
Use of noninsulin diabetes medications ^l						
Biguanide	40.9	59.9	−0.39	43.9	47.2	−0.07
Sulfonylurea	14.1	44.8	−0.72	15.9	15.2	0.02
Thiazolidinedione	4.0	2.3	0.10	5.5	2.9	0.13
DPP-4	0.7	0.9	−0.03	0.8	1.2	−0.05

(continued)

TABLE 1. (CONTINUED)

Variable	Crude (unweighted) ^a			After overlap weighting ^{a,b}		
	CGM initiators N = 149	Noninitiators N = 17,273	Standardized difference ^c	CGM initiators N = 149	Noninitiators N = 17,273	Standardized difference ^c
GLP-1	3.4	0.5	0.21	2.9	3.8	−0.05
Alpha-glucosidase inhibitor	0.7	0.2	0.08	1.0	0.2	0.10
SGLT-2	0.7	0.1	0.09	0.9	0.5	0.04
Meglitinide	0.7	0.1	0.10	0.7	0.3	0.05
Amylin	0	0.02	−0.02	0	0.4	−0.09
Number of DM therapeutic classes used						
1	54.4	28.4	0.55	50.9	46.9	0.08
2	29.5	36.5	−0.15	31.0	37.2	−0.13
3	12.8	33.3	−0.50	13.9	13.5	0.01
4+	3.4	1.9	0.09	4.2	2.4	0.10
SMBG						
Less than daily	2.0	13.0	−0.43	2.8	3.3	−0.03
At least daily	98.0	87.0	0.43	97.2	96.7	0.03
HbA1c ^m (%), mean (SD)	6.98 (0.7)	7.10 (0.6)	−0.20	6.91 (0.7)	7.02 (0.7)	−0.15
Hypoglycemia event ⁿ	0	0	0	0	0	0
Hyperglycemia event ⁿ	2.0	0.7	0.12	1.5	0.8	0.06
Any ED encounter ^o	29.5	32.3	−0.06	31.0	36.2	−0.11
Any hospitalization ^o	13.4	9.5	0.12	13.2	10.2	0.09
Number of outpatient visits ^o , mean (SD)	5.7 (4.2)	3.8 (4.0)	0.46	5.5 (4.5)	5.6 (4.4)	−0.03
Number of telephone visits ^o , mean (SD)	5.1 (6.3)	4.1 (5.7)	0.17	4.9 (6.2)	4.6 (6.2)	0.05

^aColumn percent unless otherwise specified.

^bPropensity score models were based only on the variables included in this table (Supplementary Table S1).

^cThe standardized difference (D) compares characteristics for CGM initiators with noninitiators. An absolute value of D ($|D|$) ≤ 0.1 indicates a negligible difference in the mean or in the prevalence of a covariate between groups.¹²

^dSelf-reported race/ethnicity from the electronic medical record. Mixed race means more than one of the racial/ethnic categories was selected.

^eThe neighborhood deprivation index¹⁷ is a validated contextual measure of socioeconomic status derived by linking each participant's geocoded residential address to census tract level socioeconomic indicators as reported in the 2010 American Community Survey.

^fKaiser Foundation Health Plan is the provider of all insurance and does not offer Medicare fee-for-service; insurance types are not mutually exclusive.

^gBased on the modified version of the Deyo Charlson Comorbidity Score,¹⁸ using inpatient and outpatient diagnosis and procedure codes. Possible scores ranged from 0 to 20 and represent the number of comorbid conditions identified (excluding diabetes) during the 12 months prebaseline.

^hThis validated risk stratification score¹⁹ predicts the 12-month risk of severe hypoglycemia (ED visit or hospitalization) in participants using electronic medical record data. The score is categorized as low (<1%), intermediate (1%–5%), and high (>5%).

ⁱNPH, Neutral Protamine Hagedorn; insulin types are not mutually exclusive.

^jGlucagon prescription may be an indicator of higher risk for a hypoglycemic event. Data were not available regarding unfilled prescriptions or usage.

^kPump users often use pen as a backup, so participants with both pen and pump were categorized as pump users.

^lTable reports medications at the therapeutic drug class level.

^mLast HbA1c recorded during the 12 months prebaseline. HbA1c was included in propensity score models used for the hypoglycemia outcome but not HbA1c outcomes. Therefore, it appears here both as a baseline measure and as an outcome measure. The same is true of hypoglycemic events.

ⁿIdentified by a principal diagnosis in the inpatient setting or a primary diagnosis in the ED during the 12 months prebaseline.

^oUtilization captured during the 12 months prebaseline.

CGM, continuous glucose monitoring; ED, emergency department; HbA1c, hemoglobin A1c; SMBG, self-monitoring of blood glucose.

with a decline from 35.1% to 33.2% (difference −1.9) among noninitiators, for a weighted and adjusted net change in prevalence associated with CGM initiation of 11.7% (95% CI: −1.2, 24.6; $P=0.08$). The change in the prevalence of HbA1c <8% (i.e., the baseline eligibility criteria) among CGM initiators compared with noninitiators was not significant. Fewer CGM initiators had HbA1c >9% compared with the reference (1.4% vs. 6.6%, respectively; $P=0.009$). Pre-post changes in severe hypoglycemia rates did not differ significantly between CGM initiators and noninitiators.

Discussion

CGM initiation was associated with a net improvement in glycemic control (0.30 points lower HbA1c) in this observational cohort of insulin-treated patients with T2D who used SMBG, had HbA1c <8%, and no recent severe hypoglycemia. This net benefit was largely attributable to the significant deterioration of glycemic control among noninitiators rather than improvement in glycemic control among CGM initiators. However, no changes were observed in the

TABLE 2. 12-MONTH PRE- AND POSTBASELINE DIFFERENCES IN A COHORT OF 17,422 INSULIN-TREATED PATIENTS WITH TYPE 2 DIABETES WHO USED SMBG, HAD HbA1c <8%, AND NO RECENT SEVERE HYPOGLYCEMIA

Percent (%) unless otherwise indicated	Real-time continuous glucose monitor initiators (n = 149)				Noninitiators (n = 17,273)				Difference in differences	
	Prebaseline	Postbaseline	Pre-Post difference ^a	Prebaseline	Postbaseline	Pre-Post difference ^a	Unweighted and unadjusted ^b (95% CI)	Weighted and adjusted ^c (95% CI)	P	
HbA1c (mean)	6.98	6.92	-0.06	7.10	7.42	0.32	-0.37 (-0.50, -0.24)	-0.30 (-0.50, -0.10)	0.004	
HbA1c <7%	40.3	53.7	13.4	35.1	33.2	-1.9	15.2 (6.2, 24.2)	11.7 (-1.2, 24.6)	0.08	
HbA1c <8%	100	87.8	-12.2	100	76.6	-23.4	11.2 (5.8, 16.5)	5.3 (-1.6, 12.3)	0.13	
HbA1c >9%	0	1.4	1.4	0	6.6	6.6	-5.2 (-7.1, -3.3)	-4.0 (-7.0, -1.0)	0.009	
Hypoglycemia event (ED or hospitalization)	0	3.4	3.4	0	2.06	2.06	1.30 (-1.61, 4.19)	1.70 (-1.69, 5.10)	0.33	

^aPre-post difference is calculated as postbaseline value minus prebaseline value.

^bDiD (95% CI) is calculated as ((postbaseline value minus prebaseline value in CGM initiators) minus [postbaseline value minus prebaseline value in CGM noninitiators]) from repeated-measures generalized linear models; there are slight discrepancies between the unadjusted/unweighted model-based DiD and the crude DiD for the HbA_{1c} outcomes due to missing data. Patients with ≥ 1 HbA_{1c} measure in either the pre- or postbaseline were included in the model.

^cOverlap weighting was based on propensity score model and adjusted for variables identified in the directed acyclic graph as critical adjusters (prebaseline insulin treatment, glycemic control, and acute metabolic crisis) and baseline covariates with $|D| > 0.10$ after weighting. Conservative 95% confidence intervals for both DiD estimates were estimated using robust variance estimators.⁷

DiD, difference-in-differences.

risk of severe hypoglycemia. Although rigorous causal estimation methods achieved excellent balance in covariates,³ these observational findings may have been susceptible to selection bias or regression to the mean.

Observed patterns are unlikely due to treatment intensification in the CGM initiators or lack thereof in the non-initiators since mean HbA_{1c} after baseline was still <8% (i.e., below the glycemic threshold indicating an urgent need for intensification). CGM initiation may have enabled patients to better manage their insulin and health behaviors than SMBG alone. Additional observational studies from other health care settings or randomized trials of CGM in well-controlled patients with T2D should be conducted to confirm these observational findings.

Conclusions

This real-world comparative effectiveness study suggests that CGM may provide benefit by preventing glycemic deterioration even in well-controlled patients with insulin-treated T2D.

Disclaimer

The sponsors had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the article; and had no role in the decision to submit the article for publication.

Authors' Contributions

A.J.K., M.M.P., H.H.M., L.K.G., and R.D. were involved in study design, researched data, contributed to discussion, wrote/edited the article, and reviewed/edited the article. M.M.P. performed statistical analyses.

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Supplementary Material

Supplementary Table S1

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